Sum of first 13 terms $= 234 \quad \Rightarrow \; S_{13} = \frac{13}{2}[2a + 12d] \; = 234$

$$2a + 12d \; = 36 \qquad \ldots(2)$$

Solving (1) and (2) we get, $a = 33,\ d = \frac{-5}{2}$

Therefore, common difference is $\frac{-5}{2}$.

Sum of first 21 terms $S_{21} = \frac{21}{2}\left[2 \times 33 + (21-1) \times \left(-\frac{5}{2}\right)\right] = \frac{21}{2}[66 - 50] = 168.$

Example 2.35 In an A.P. the sum of first n terms is $\frac{5n^2}{2} + \frac{3n}{2}$. Find the 17th term.

Solution The 17th term can be obtained by subtracting the sum of first 16 terms from the sum of first 17 terms

$$S_{17} = \frac{5 \times (17)^2}{2} + \frac{3 \times 17}{2} = \frac{1445}{2} + \frac{51}{2} = 748$$

$$S_{16} = \frac{5 \times (16)^2}{2} + \frac{3 \times 16}{2} = \frac{1280}{2} + \frac{48}{2} = 664$$

Now, $\quad t_{17} = S_{17} - S_{16} = 748 - 664 = 84$

Example 2.36 Find the sum of all natural numbers between 300 and 600 which are divisible by 7.

Solution The natural numbers between 300 and 600 which are divisible by 7 are 301, 308, 315, …, 595.

The sum of all natural numbers between 300 and 600 is $301 + 308 + 315 + \cdots + 595$.

The terms of the above series are in A.P.

First term $a = 301$; common difference $d = 7$; Last term $l = 595$.

$$n = \left(\frac{l - a}{d}\right) + 1 = \left(\frac{595 - 301}{7}\right) + 1 = 43$$

$\because S_n = \frac{n}{2}[a + l]$, we have $S_{43} = \frac{43}{2}[301 + 595] = 19264.$

Example 2.37 A mosaic is designed in the shape of an equilateral triangle, 12 ft on each side. Each tile in the mosaic is in the shape of an equilateral triangle of 12 inch side. The tiles are alternate in colour as shown in the figure. Find the number of tiles of each colour and total number of tiles in the mosaic.

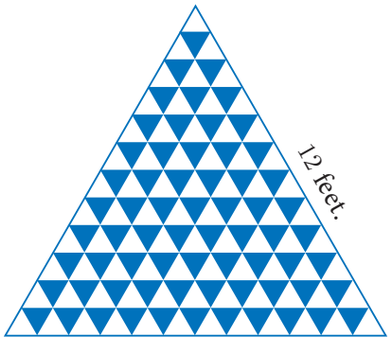


Fig.2.12

Solution Since the mosaic is in the shape of an equilateral triangle of 12 feet, and the tile is in the shape of an equilateral triangle of 12 inch (1 feet), there will be 12 rows in the mosaic.

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